Learning to Insert: Image-Grounded Cell Recovery from Incomplete Table Structure Predictions

Project Proposal

October 2026

Abstract

Table Structure Recognition (TSR) systems usually predict a set of table cells and then convert these predictions into a final logical structure. A recurring failure is an omitted cell instance: the image may contain a valid cell, but the detector does not provide a usable candidate for it. A later selector can choose among existing candidates, but it cannot select a candidate that is absent. This proposal studies whether a TSR model can explicitly expand an incomplete candidate set by inserting new cell hypotheses from the table image and the surviving detected cells.

The project is deliberately narrower than general table reconstruction. The proposed task is *detection-conditioned cell insertion*: given a table image and an incomplete set of cell candidates produced by a base detector, predict whether an additional cell should be inserted and, when appropriate, estimate its box and logical span. The insertion decoder must also support a no-insertion outcome and avoid subdividing a valid spanning cell. The first implementation will attach a lightweight recovery decoder to a fixed cell detector, with a set or graph encoder for surviving-cell context and image-attention queries for candidate generation. Training will start with controlled deletion of correct candidates and will then move to natural detector omissions. Evaluation will distinguish cells that were merely filtered from cells absent even from a generous pre-filter candidate pool.

The literature review shows that missing-cell reconstruction, vacancy detection, topology-aware refinement, and joint spatial-logical prediction have all been studied. The proposed contribution is therefore not "missing-cell reconstruction" in general. The narrower question is whether a learned, variable-cardinality insertion operator can expand an incomplete detector-produced candidate set using raster evidence and surviving-cell context, while explicitly learning when *not* to insert. The core evidence required is recovery of natural omissions at matched insertion precision, with ablations showing that both image evidence and surviving-cell context contribute.

1 Research Objective

1.1 Problem definition

Let a table image be I. A base TSR detector produces an initial candidate set

$$C_0 = G_\theta(I), \tag{1}$$

where each candidate may contain a bounding box, confidence score, and, depending on the host detector, logical information.

Let C^* denote the ground-truth set of cell instances. In many cases,

$$C^* \setminus C_0 \neq \emptyset, \tag{2}$$

so at least one valid cell has no usable representation in the initial candidate set. We call this **candidate-set incompleteness**.

The proposed recovery module receives the image and the surviving candidates:

$$C_{\text{ins}} = R_\phi(I, C_0), \tag{3}$$

and forms

$$C_1 = C_0 \cup C_{\text{ins}}. \tag{4}$$

Each inserted candidate may be represented as

$$c = (b, p_{\text{row}}, p_{\text{col}}, s_{\text{row}}, s_{\text{col}}), \tag{5}$$

where $b = (x, y, w, h)$ is the cell box, p_{row} and p_{col} denote logical row and column attachment, and $s_{\text{row}}, s_{\text{col}}$ denote row and column spans.

The decision space is intentionally broader than "insert a box whenever a geometric gap exists":

$$z \in \{\texttt{insert}, \texttt{no-insert}, \texttt{span-compatible}\}. \tag{6}$$

The last outcome represents an apparent vacancy that should not be turned into an independent cell because it is explained by an existing spanning structure.

1.2 Central research question

> Can a learned insertion operator recover cell instances that are absent from an incomplete detector candidate set, using surviving-cell context and raster-image evidence, without introducing unsupported cells or breaking valid spans?

1.3 Research hypotheses

1. **H1 - Candidate incompleteness:** A non-trivial fraction of TSR cell errors can be traced to cells that are missing from the retained candidate set.
2. **H2 - Threshold relaxation is insufficient:** Lowering the detector threshold can increase candidate recall, but the resulting false positives and duplicates prevent it from being an equivalent solution to learned insertion.
3. **H3 - Learned insertion can recover natural omissions:** A recovery decoder can generate correct cell instances that are absent even from a generous pre-filter candidate pool.
4. **H4 - Two sources of evidence matter:** Surviving-cell context and local image evidence each contribute to insertion accuracy; removing either should reduce performance.
5. **H5 - Explicit no-insertion matters:** A model trained to decide *whether* to insert, rather than only to predict a box, should reduce false insertions and erroneous subdivision of spanning cells.

2 Literature Survey

The project sits at the intersection of cell detection, structural relation modeling, table reconstruction, and recent image-to-structure models. The important distinction is between methods that improve the original detector, methods that assemble a full structure from predicted elements, and methods that explicitly create additional cell instances.

2.1 Early cell and structure detection

Early TSR systems approached table structure as either a detection problem or a decomposition problem. DeepDeSRT used deep detection and structure recognition for document tables [1]. CascadeTabNet combined table detection and cell recognition in a CNN-based pipeline [2]. GraphTSR framed TSR as graph relation prediction between detected table cells and established a graph-based formulation for complex scientific tables [3]. These methods demonstrate that cell localization alone does not determine the final table structure; relations among cells are needed.

2.2 Split-and-merge and grid-based methods

Split-and-merge systems predict a regular representation and subsequently merge elements to account for spanning cells. Deep Splitting and Merging [4] and SEM (Split, Embed and Merge) [5] are representative examples. TSRFormer models table separators with a Transformer and reconstructs spans from the predicted grid [9]. GridFormer uses grid prediction and structural relations to model the table layout [10].

These methods are relevant because a regular grid can expose apparent vacancies. However, the existence of a geometric vacancy is not enough to establish that an independent cell should be inserted. A vacancy may be an empty region belonging to a spanning cell or may arise from an imperfect detection.

2.3 Global structure and learned reconstruction

LGPMA (Local and Global Pyramid Mask Alignment) is particularly relevant to the proposed problem [6]. It predicts local cell-related information together with global table segmentation and includes an empty-cell recovery process based on structural alignment. Its recovery procedure searches for vacant positions and uses image-level segmentation to determine how regions should be merged.

LGPMA therefore rules out a broad novelty claim of "combining topology with image evidence for missing-cell recovery." The proposed work needs to address a different intervention: learned insertion of omitted *cell instances* into an incomplete detector-produced candidate set, including a learned decision not to insert.

Relation and logical-coordinate methods further emphasize structural reasoning. TGRNet jointly predicts spatial and logical locations and reconstructs the table graph [7]. LORE uses an anchor-free detection formulation and predicts logical coordinates alongside spatial boxes [13]. TableCenterNet similarly targets one-stage prediction of spatial and logical cell locations [15]. These methods strengthen the detector or predict structure jointly; they do not, by themselves, establish an insertion operator whose output cardinality can exceed an incomplete candidate set.

2.4 Sequence generation and image-to-structure models

TableFormer uses Transformer decoding for table structure understanding and combines cell localization with structured sequence generation [11]. VAST introduces visual-alignment sequential coordinate modeling and uses visual alignment to improve coordinate prediction [14]. More recent models extend this trend. TableSeq represents image, structure, and cell localization in a unified autoregressive sequence [23]. FastTab uses recursive structure prediction and ROI-aligned features for grid and span inference [24]. StrucTab introduces intermediate structural supervision and a structured optimization framework for table parsing [25]. BGTR uses detected bounding

boxes to guide HTML generation [26]. TPGCAT introduces text-prior-guided cell alignment for table structure recognition [27].

These models show that strong TSR systems can reason over geometry and structure jointly. They do not make the specific problem of expanding an incomplete candidate set the primary inference operation.

2.5 Missing-cell and reconstruction methods

The closest direct prior work is a line of table reconstruction methods that infer cells absent from an initial detection. The historical-record reconstruction work published in IJDAR in 2025 explicitly infers missing cells from the horizontal and vertical alignment of detected positive cells and generates candidates at identified gaps [16]. A related 2025 IJDAR study on semi-supervised TSR also evaluates missing-cell generation and reconstruction in historical tables [17].

These systems establish that coordinate-based candidate generation from surviving detections is not new. They motivate using such geometric completion as a required baseline rather than as the proposed method.

TableFormer also contains PDF-oriented handling of orphan content that can lead to structural cell creation from document geometry [11]. This is not equivalent to the proposed raster-only inference setting because PDF text and geometry provide information that is unavailable to a pure image-based model. It is nevertheless evidence that structural cell creation itself is not a new operation.

2.6 Detector failure and suppression

Detection-based TSR systems also expose a different route to omission: the detector can generate a weak prediction that is later removed. Table Transformer (TATR) uses DETR-style object detection for cells, rows, and columns in the PubTables-1M benchmark [12]. TableCenterNet, LORE, and other detector families similarly rely on confidence scores or peak selection. Recent work such as OG-HFYOLO focuses on improving table-cell instance detection and reducing omissions or suppression errors [21]. ConRTF targets boundary localization through an edge-constrained training objective rather than a candidate insertion stage [22].

These methods suggest an important experimental control: any learned recovery method must be compared against simply lowering the detector threshold or improving the detector itself.

2.7 Scene and deformed tables

WTW ("Parsing Table Structures in the Wild") was introduced for tables with natural visual deformation, including inclined, curved, blurred or occluded, extreme-aspect-ratio, overlaid, multicolor, and irregular tables [8]. Methods such as SCAN address inclined and curved tables using segmentation collaboration and cell alignment [20]. WTW is therefore useful as a stress test for whether candidate insertion remains valid when a regular-grid assumption becomes unreliable.

3 What Is Actually Novel?

After the literature survey, the novelty claim must be narrow. The following statements are already established and should not be used as the main contribution:

Table 1: Novelty boundary after the literature survey.

Claim	Assessment
Recover missing cells from surviving detections	Established in geometric and historical-record reconstruction.
Find vacancies using row/column alignment or topology	Established.
Use image evidence together with structural information	Established by methods such as LGPMA and several modern TSR models.
Refine cell geometry with structural context	Established.
Create new structural cells in a reconstruction pipeline	Established in several settings, including PDF-oriented processing.
Learned insertion of omitted instances from an incomplete detector set, with variable output cardinality	**Narrow contribution to test.**
Explicit insert / no-insert / span-compatible decision	**Potentially differentiating; requires experiments.**
Training under candidate deletion and evaluation on natural omissions, including cells absent from a generous candidate pool	**A useful evaluation and training package.**

The proposed contribution is therefore not a new form of geometric gap filling. It is a learned operator that takes an incomplete set of detector-produced instances and decides whether additional instances are justified by the image and the surviving table structure. The output cardinality is allowed to grow, and the model can explicitly output "no insertion."

A proposed formal statement is:

> We study detection-conditioned cell insertion for TSR: given a raster table image and an incomplete candidate set produced by a base cell detector, learn a variable-cardinality decoder that inserts additional cell instances only when image evidence and surviving-cell context support them, while preserving valid spanning structures.

This claim is deliberately weaker than a "first missing-cell recovery" claim. A final first-of-its-kind statement should only be made after an exhaustive search immediately before submission.

4 Proposed Framework

4.1 Design principles

The first implementation should satisfy five constraints:

1. The base detector is kept fixed when measuring the contribution of recovery.
2. The recovery module operates on raster image features and detector outputs; OCR or PDF metadata are not required at inference.
3. The recovery decoder has a variable number of outputs rather than one fixed query per geometric gap.

4. A candidate may be classified as "no insertion," which prevents the model from turning every vacancy into a cell.
5. Existing detector predictions are initially left unchanged so that the experiment isolates the value of candidate insertion.

4.2 Initial host detector

The first host should be Table Transformer (TATR) because it has an established implementation, is trained for cell/row/column detection on PubTables-1M, and exposes explicit cell boxes and confidence scores [12]. The same recovery module can later be attached to LORE or TableCenterNet to test whether the method is tied to one detector family [13, 15].

4.3 Architecture

The proposed initial architecture, LEARN2INSERT, contains three learned components around a frozen or jointly fine-tuned host detector.

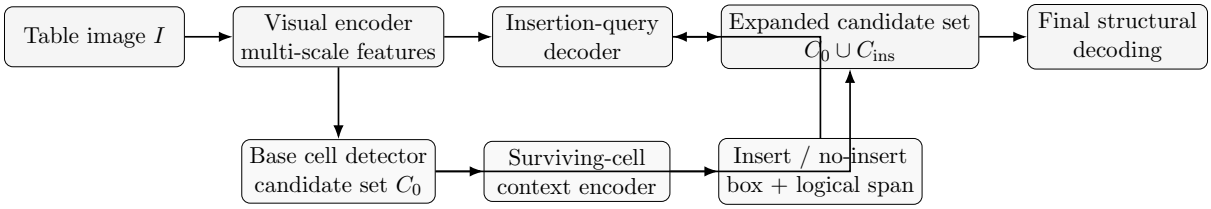


Figure 1: Initial framework. The base detector provides surviving cell candidates, while the insertion decoder combines their context with image features and can produce a variable number of new candidates.

4.3.1 Visual encoder

A ResNet-50/FPN or a similar moderate-size image encoder is sufficient for the first prototype. The aim is not to introduce a new backbone. Features $F = E(I)$ are sampled around candidate and query regions.

4.3.2 Surviving-cell context encoder

Each retained candidate becomes a node with feature vector

$$h_i = [h_i^{\text{vis}}; h_i^{\text{geom}}; s_i], \tag{7}$$

where h_i^{vis} is an ROI feature, h_i^{geom} encodes normalized box coordinates and size, and s_i is detector confidence.

Edges or attention biases can encode relative horizontal/vertical distance, alignment, overlap, and containment. A small graph network or set Transformer produces contextual representations

$$H = \text{ContextEncoder}(\{h_i\}_{i=1}^{N}). \tag{8}$$

4.3.3 Insertion-query decoder

Use a small learned query bank $Q = \{q_j\}_{j=1}^{M}$, rather than constructing one query for every geometric vacancy. Each query attends to both H and visual features F:

$$z_j = \mathrm{CrossAttn}(q_j, H, F). \tag{9}$$

The decoder produces

$$p_j^{\mathrm{ins}} = \sigma(f_{\mathrm{cls}}(z_j)), \tag{10}$$

$$\hat{b}_j = f_{\mathrm{box}}(z_j), \tag{11}$$

$$\hat{r}_j = f_{\mathrm{row}}(z_j), \tag{12}$$

$$\hat{c}_j = f_{\mathrm{col}}(z_j), \tag{13}$$

$$\hat{s}_j = f_{\mathrm{span}}(z_j). \tag{14}$$

A "no insertion" state is represented by the classification head. Span prediction can be represented either by start/end row and column offsets or by discrete span classes in the first implementation.

4.3.4 Compatibility and duplicate suppression

For an inserted candidate c_j, a compatibility head estimates whether the proposal is geometrically and structurally compatible with nearby surviving cells. The first prototype should use deterministic duplicate removal after insertion. A learned compatibility score can be added later if the simple version produces useful recovery gains.

5 Training Strategy

5.1 Controlled candidate deletion

The first training stage creates incomplete candidate sets from annotated tables. Let C^+ be the subset of ground-truth cells represented by correctly matched base-detector candidates. A mask $M \subset C^+$ is sampled and removed from the input candidate set:

$$C_0' = C_0 \setminus M. \tag{15}$$

The image remains unchanged. The recovery decoder is trained to produce candidates matching the removed cells.

Deletion should not be purely random. Four cases should be included:

1. isolated cell deletion;
2. adjacent deletion of two or more cells;
3. deletion near a spanning cell;
4. deletion patterns sampled from real detector errors.

The last case is introduced after a baseline detector has been profiled.

5.2 Set-based supervision

Because the number of insertion queries is fixed while the number of missing cells varies, a DETR-style matching loss is suitable. For matched prediction-target pairs:

$$\mathcal{L}_{\mathrm{ins}} = \lambda_{\mathrm{cls}}\mathcal{L}_{\mathrm{cls}} + \lambda_{\mathrm{box}}\mathcal{L}_{\mathrm{box}} + \lambda_{\mathrm{span}}\mathcal{L}_{\mathrm{span}}. \tag{16}$$

The box term may combine L1 loss and generalized IoU loss:

$$\mathcal{L}_{\text{box}} = \alpha \|b - \hat{b}\|_1 + \beta \left(1 - \text{GIoU}(b, \hat{b})\right). \tag{17}$$

Unmatched queries are supervised as "no insertion." This is important because the method should not be rewarded merely for increasing the number of candidates.

5.3 Transition to natural omissions

After the synthetic deletion stage converges, the training data should include candidate sets produced by the real detector. For each image, the recovery module sees exactly the candidates that the detector would normally retain. Ground truth is used only to define targets during training. At test time the module receives no deletion mask and no ground-truth topology.

6 Experimental Design

6.1 Evaluation protocol 1: controlled omissions

Start from detector outputs and deliberately remove selected true-positive candidates while leaving the original image untouched. This isolates the insertion capability from the detector's unrelated errors.

Evaluate deletion rates of 10%, 20%, 30%, 40%, and 50% of eligible candidates. The evaluation should contain isolated deletions, adjacent deletions, and deletions close to spanning cells. Intact tables with no deleted candidates are also required. On those tables, the correct behavior is often "no insertion."

6.2 Evaluation protocol 2: natural omissions

Run the base detector normally. After inference, match its candidates to the ground-truth cells only for analysis. Split the missed cells into two groups:

1. **filtered omissions**: a corresponding candidate exists in a pre-filter or low-confidence pool but is removed by the standard threshold;
2. **proposal omissions**: no candidate in a generous pre-filter pool matches the ground-truth cell.

The second category is the strongest test of the proposed method because the recovery module must create an instance rather than recover a discarded low-confidence prediction.

6.3 Baselines

The central comparison should keep the host detector fixed.

For whole-system context, report LGPMA, SEM, TableFormer, LORE, TableCenterNet, and representative recent TSR systems where reproducible implementations and compatible evaluation protocols are available. These comparisons should be presented separately from the same-detector intervention table so that a change in backbone or decoding paradigm is not confused with the effect of insertion.

Table 2: Same-detector baselines and the question each one answers.

Variant	Purpose
Base detector	Establishes the initial candidate and final structure quality.
Lower detection threshold	Tests whether retaining more detector candidates is sufficient.
Geometric gap completion	Tests coordinate-based reconstruction from surviving cells.
Topology/context-only recovery	Tests whether visual evidence is necessary.
Image-only recovery	Tests whether surviving-cell context is necessary.
Full recovery model	Uses both candidate context and image evidence.

6.4 Datasets

PubTables-1M contains approximately 947,642 tables and provides large-scale cell, row, and column annotations, including complex tables with spanning cells [12]. SciTSR contains 15,000 scientific tables, with a designated complex subset (SciTSR-COMP) in the test set [28]. WTW contains 14,581 table images and covers several types of natural visual deformation; its annotations include quadrilateral cell geometry and logical row/column coordinates [8].

6.5 Metrics

The evaluation has two layers: standard TSR metrics and task-specific insertion metrics.

6.5.1 Standard TSR metrics

- **GriTS:** report topology, location, and content components where the dataset supports them. GriTS was designed to compare table structures as grids and separates topology and localization effects [18].
- **TEDS:** use for tree-structure similarity when an HTML-like representation is available [19].
- **Cell localization:** precision, recall, F1, AP_{50}, AP_{75}, and average recall where appropriate.
- **SciTSR relation accuracy:** report the standard cell adjacency relation precision, recall, and F1 used for the dataset [28].
- **WTW structure/localization:** follow the official evaluation protocol and include logical structure plus cell localization metrics; TEDS can be reported where the conversion to the required tree format is reliable.

6.5.2 Candidate-recovery metrics

For the new intervention, define a prediction as a matched insertion when its box reaches a chosen IoU threshold with an omitted ground-truth cell and its logical attachment is compatible.

Insertion precision

$$P_{\text{ins}} = \frac{|C_{\text{ins}} \cap C_{\text{miss}}|}{|C_{\text{ins}}|}. \tag{18}$$

Recovery recall

$$R_{\text{rec}} = \frac{|C_{\text{ins}} \cap C_{\text{miss}}|}{|C_{\text{miss}}|}. \tag{19}$$

Table 3: Planned datasets and their roles.

Dataset	Role	Protocol
PubTables-1M	Main development and in-domain benchmark.	Train the host/recovery models, tune thresholds on validation, report primary candidate and structure results on the official test split.
SciTSR	Cross-dataset structural evaluation.	Transfer the trained recovery model where compatible; optionally fine-tune on the training split as a separate setting. Report full test performance.
SciTSR-COMP	Stress test for complex scientific tables.	Report separately as a subset of the SciTSR test set; focus on spans and dense structural relations.
WTW	Cross-domain test for deformed and scene tables.	Evaluate natural deformation, irregular geometry, and logical cell reconstruction.
ICDAR-2013	Optional legacy comparison.	Use as a small external sanity check rather than the main evidence.
PubTabNet	Optional auxiliary benchmark.	Add only after verifying the chosen cell-level and structure annotations support the insertion metrics.
PubTables-v2	Optional extension.	Consider later for multi-page/full-page settings once the single-page insertion task is established.

Insertion F1

$$F1_{\text{ins}} = \frac{2P_{\text{ins}}R_{\text{rec}}}{P_{\text{ins}} + R_{\text{rec}}}. \tag{20}$$

False insertion rate

$$FIR = \frac{|C_{\text{ins}} \setminus C^*|}{|C_{\text{ins}}|}. \tag{21}$$

Candidate recall gain

$$\Delta CR = CR(C_0 \cup C_{\text{ins}}) - CR(C_0), \tag{22}$$

where CR is computed against the cell ground truth under the selected matching protocol.

Span violation rate Measure the fraction of inserted cells that create a subdivision inside a ground-truth spanning cell or that conflict with a valid span assignment.

No-change accuracy On intact tables, report the fraction of tables for which the recovery module inserts no unsupported cell.

All insertion metrics should be reported at more than one IoU threshold, with 0.50 and 0.75 as the initial settings. A stricter structural match can require exact start/end row and column coordinates when those annotations exist.

7 Ablation Plan

The key ablation is a 2×2 factorization of information sources:

Table 4: Core information ablation.

	No candidate context	**Candidate context**
No image evidence	No recovery / control	Context-only
Image evidence	Image-only	**Full model**

Additional ablations should test:

1. number of insertion queries M;
2. deletion ratio during training;
3. random versus detector-derived deletion;
4. box-only versus box-plus-span prediction;
5. with versus without the no-insertion class;
6. deterministic versus learned compatibility filtering;
7. one-pass versus iterative recovery, only after the one-pass model is understood.

The model should not be declared useful from a synthetic deletion experiment alone. Natural omissions are the deciding evaluation.

8 Failure Analysis

A central part of the project is to record the failure modes created by insertion. At minimum, each false positive should be assigned to one of the following categories:

1. false geometric gap;
2. over-segmentation of a spanning cell;
3. duplicate of an existing detector candidate;
4. incorrect box despite correct structural location;
5. incorrect logical row/column attachment;
6. unsupported insertion on an intact table.

For true recoveries, record whether the cell was filtered from a low-confidence candidate pool or absent from the generous pre-filter pool. The second case should be reported as a separate headline result.

9 Computational Scope and Feasibility

The initial system is intentionally modest. The first version does not require a large vision-language model, OCR generation, reinforcement learning, or an exact optimizer. The host detector and visual encoder can be kept frozen while the recovery module is trained. This makes the first study feasible on a single consumer GPU with staged experiments.

A practical sequence is:

1. build the candidate-tracing and controlled-deletion pipeline;
2. implement geometric completion and lower-threshold baselines;
3. train an image-only insertion decoder;
4. add surviving-cell context;
5. evaluate natural omissions;
6. only then consider joint fine-tuning or a second host detector.

The first success criterion should be a controlled recovery result. The second is a natural-omission result. End-to-end GriTS/TEDS gains come only after those two results are established.

10 Expected Research Contributions

The project will aim to deliver three concrete outputs.

1. **A problem formulation:** detection-conditioned cell insertion as a distinct stage between initial cell prediction and final structural decoding.
2. **A learned insertion architecture:** a variable-cardinality decoder that uses image evidence and surviving-cell context and can output "no insertion."
3. **An evaluation protocol:** controlled candidate deletion and natural-omission analysis that separates threshold rescue from genuine proposal generation.

The central empirical claim, if supported, will be:

> At comparable insertion precision, a learned insertion module recovers more genuine detector omissions than threshold relaxation and geometric gap completion, including cells absent from a generous pre-filter candidate pool, while preserving valid spanning structures.

This is the result that would justify a full TSR architecture paper. A small gain in a final table metric without this evidence would not be sufficient.

11 Risks and Decision Criteria

A simple go/no-go rule is recommended:

> Proceed to a larger architecture only if the full model beats geometric completion and image-only recovery on natural omissions at a matched insertion-precision operating point, and the gain includes cases absent from the generous pre-filter pool.

12 Phased Work Plan

13 Positioning for a Future Paper

A future paper should avoid presenting the work as a general solution to "missing cells." The literature already contains vacancy inference, empty-cell reconstruction, structure-guided

Table 5: Main technical risks and the corresponding diagnostic test.

Risk	**Diagnostic**
Threshold relaxation solves most omissions	Compare at matched insertion precision and candidate budget.
Geometric completion is equally strong	Use the same detector output and test complex/span cases.
Image evidence is unnecessary	Compare full model to context-only recovery.
Surviving-cell context is unnecessary	Compare full model to image-only recovery.
Synthetic deletion does not resemble detector errors	Train/evaluate on natural omissions and analyze omission types.
The method inserts too many false cells	Measure false insertion rate, intact-table no-change accuracy, and span violation rate.
The improvement is host-specific	Attach the recovery module to a second detector such as LORE or TableCenterNet.

Table 6: Proposed development schedule.

Phase	**Work**	**Deliverable**
1	Reproduce host detector, build candidate matching, threshold sweeps, and omission taxonomy.	Baseline candidate-tracing report.
2	Implement controlled deletion and geometric completion.	Recovery benchmark and geometric baseline.
3	Implement image-only insertion decoder.	First learned recovery model.
4	Add surviving-cell context encoder and no-insertion/span heads.	Full first architecture.
5	Natural omission evaluation on PubTables-1M.	Main insertion results.
6	SciTSR, SciTSR-COMP, and WTW evaluation.	Cross-domain and structural stress tests.
7	Second host detector, efficiency analysis, and final ablations.	Architecture robustness study.

coordinate prediction, and modern joint TSR architectures. The paper should instead center on the following sequence:

$$\begin{aligned} &\text{incomplete detector output} \rightarrow \text{learned insertion decision} \\ &\quad \rightarrow \text{new cell candidates} \rightarrow \text{structural decoding.} \end{aligned} \tag{23}$$

The main figure should show examples in which the base detector misses a cell and the proposed module inserts it, together with hard negatives where the apparent gap belongs to a spanning cell. The main quantitative table should report insertion precision/recall and final structure metrics side by side. The most convincing qualitative examples are likely to be those where lower thresholds either fail to recover the cell or create a competing false candidate.

The project should also retain a clear distinction between a methodological claim and an empirical result. "We propose a learned insertion operator" is a design statement. "The operator recovers 18.4% of natural proposal omissions at 92.1% insertion precision" would be an empirical statement and should only appear after measurement.

14 Conclusion

The literature does not leave a broad "missing-cell reconstruction" problem untouched. LGPMA already combines structural vacancy reasoning with image-level information, historical-table work generates missing cells from aligned detections, and recent TSR models increasingly combine geometry and structure. The viable research question is narrower: whether a dedicated learned insertion stage can expand an incomplete detector candidate set in a controlled way.

The initial architecture therefore remains deliberately small: a standard cell detector, a surviving-cell context encoder, and a learned insertion decoder that attends to image features and can output either a new cell or no insertion. The experimental design is more important than architectural complexity. PubTables-1M will support the main ablations, SciTSR and SciTSR-COMP will test dense structural layouts, and WTW will test deformed real-world tables. The decisive evidence will be recovery of natural omissions that are absent from a generous candidate pool, at a precision level that threshold relaxation and geometric completion cannot match.

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This proposal distinguishes established components from the narrower contribution being tested. Any final novelty claim should be updated after a submission-time literature search.